Problem 1: Show that adding for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, adding two rows or two columns makes no difference to the determinant.

Solution

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det(A) = ad - bc$.

Adding one row to another row.

$$A' = \begin{bmatrix} a & b \\ c+a & d+b \end{bmatrix}$$

$$\det(A') = a(d+b) - b(c+a) = ad + ab - bc - ba = ad - bc = \det(A)$$

Adding one column to another column.

$$A'' = \begin{bmatrix} a & b+a \\ c & d+c \end{bmatrix}$$

$$\det(A'') = a(d+c) - c(b+a) = ad + ac - bc - ca = ad - bc = \det(A)$$

Problem 2: If A is an $n \times n$ matrix, and A is invertible, then prove that $AX = 0$ has only trivial solution.

Solution

Consider the equation $AX = 0$. Multiply both sides of the equation by A^{-1} on the left:

$$A^{-1}(AX) = A^{-1}0$$

$$(A^{-1}A)X = 0$$

$$IX = 0$$

$$X = 0$$

Therefore, the only solution to $AX = 0$ is the trivial solution $X = 0$.

Problem 3: Show that if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ then:

$$\det(AB) = \det(A) \cdot \det(B)$$

Solution

$AB = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$.

$\det(A) = ad - bc$
$\det(B) = eh - fg$

$$\begin{aligned} \det(AB) &= (ae+bg)(cf+dh) - (af+bh)(ce+dg) \\ &= aecf + aedh + bgcf + bgdh - afce - afdg - bhce - bhdg \\ &= aedh + bgcf - afdg - bhce \\ &= adeh - adfg + bceh - bcgf \\ &= ad(eh - fg) - bc(eh - fg) \\ &= (ad - bc)(eh - fg) \\ &= \det(A)\det(B) \end{aligned}$$

Therefore, $\det(AB) = \det(A)\det(B)$.

Problem 4: If A is an invertible $n \times n$ matrix, then prove that for each $n \times 1$ matrix B, the system of equations $AX = B$ has exactly one solution $X = A^{-1}B$.

Solution

Existence:
Suppose $X = A^{-1}B$. Then, substituting this into the equation $AX = B$:

$$AX = A(A^{-1}B) = (AA^{-1})B = IB = B$$

Thus, $X = A^{-1}B$ is a solution to the system $AX = B$.

Uniqueness:
Suppose there are two solutions, X_1 and X_2, to the system $AX = B$. Then:

$$AX_1 = B \text{ and } AX_2 = B$$

Subtracting the two equations:

$$AX_1 - AX_2 = B - B \implies A(X_1 - X_2) = 0$$

Since A is invertible, we can multiply both sides by A^{-1} on the left:

$$A^{-1}A(X_1 - X_2) = A^{-1}0 \implies I(X_1 - X_2) = 0 \implies X_1 - X_2 = 0 \implies X_1 = X_2$$

Therefore, the solution is unique.

Problem 5: Compute the determinant of the matrix $A = \begin{bmatrix} a & b & c \\ 5 & 5 & 5 \\ b+c & a+c & a+b \end{bmatrix}$.

Solution

The determinant doesn't change by adding rows. We add row 1 to row 3.

$$\begin{vmatrix} a & b & c \\ 5 & 5 & 5 \\ b+c & a+c & a+b \end{vmatrix} = \begin{vmatrix} a & b & c \\ 5 & 5 & 5 \\ a+b+c & a+b+c & a+b+c \end{vmatrix} = 5(a+b+c)\begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

Having two rows identical makes the determinant zero. Hence,

$\det(A) = 5(a+b+c) \times 0 = 0.$

Alternatively, we can compute the determinant of A by expanding from first row.

$$\begin{aligned} \det(A) &= a\begin{vmatrix} 5 & 5 \\ a+c & a+b \end{vmatrix} - b\begin{vmatrix} 5 & 5 \\ b+c & a+b \end{vmatrix} + c\begin{vmatrix} 5 & 5 \\ b+c & a+c \end{vmatrix} \\ &= a[5(a+b) - 5(a+c)] - b[5(a+b) - 5(b+c)] + c[5(a+c) - 5(b+c)] \\ &= a[5a + 5b - 5a - 5c] - b[5a + 5b - 5b - 5c] + c[5a + 5c - 5b - 5c] \\ &= a(5b - 5c) - b(5a - 5c) + c(5a - 5b) \\ &= 5ab - 5ac - 5ab + 5bc + 5ac - 5bc \\ &= 0 \end{aligned}$$

Therefore, the determinant of the given matrix is 0.

Problem 6: Let $AX = B$ be the system of equation and A is a square matrix, what can we say about the solution if we know the determinant of A is zero? and nonzero?

Solution

Case 1: det(A) $\neq$ 0
If the determinant of A is non-zero, then A is invertible. This means there exists a unique solution to the system $AX = B$, given by $X = A^{-1}B$. The system is consistent, meaning a solution exists, and the solution is unique. (Proof is given in Problem4.)
Case 2: det(A) = 0
If the determinant of A is zero, then A is not invertible. In this case, the system $AX = B$ can have either no solutions or infinitely many solutions. We cannot say for certain which is the case without further analysis.
No solutions: The system is inconsistent. This occurs when the augmented matrix $[A|B]$ has a row of the form $[0\ 0\ \ldots\ 0 \mid b]$ where $b \neq 0$.
Infinitely many solutions: The system is consistent, but the solutions are not unique. This occurs when the augmented matrix $[A|B]$ can be reduced to a form with at least one free variable (a variable that is not a leading variable in any row). The presence of free variables implies that some variables can take on any value, leading to infinitely many solutions.

Problem 7: Compute the determinant of the matrix $\begin{bmatrix} 2 & 2 & 2 \\ x & y & z \\ y+z & x+z & x+y \end{bmatrix}$.

Solution

Same as Problem 5.

Problem 8: Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ and it is given that $\det(A) = -10$. Then find $\det(5A^{-1})$.

Solution

We know that $\det(kA) = k^n \det(A)$ for an $n \times n$ matrix A. Also, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Since A is a 3×3 matrix, A^{-1} is also a 3×3 matrix. Therefore, $5A^{-1}$ is also a 3×3 matrix.

$\det(5A^{-1}) = 5^3 \det(A^{-1}) = \frac{5^3}{\det(A)} = -\frac{125}{10} = -\frac{25}{2}$

Problem 9: If A is an $n \times n$ matrix and $AX = 0$ has only trivial solution, then prove that the reduced row-echelon form of A is I_n, where $AX = 0$ is the matrix form of the system:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= 0 \end{aligned}$$

Solution

Refer to the book.

Problem 10: Prove that if A is an invertible square matrix, then:

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Solution

By the definition of the inverse of a matrix, we have $AA^{-1} = I$,where I is the $n \times n$ identity matrix.

$$\begin{aligned} \det(AA^{-1}) &= \det(I) \\ \implies \det(A)\det(A^{-1}) &= 1 \\ \implies \det(A^{-1}) &= \frac{1}{\det(A)} \end{aligned}$$

Problem 11: If A is an $n \times n$ matrix, and the reduced row-echelon form of A is I_n, then prove that A is expressible as a product of elementary matrices.

Solution

The reduced row-echelon form of A is obtained by a sequence of elementary row operations. Each elementary row operation can be represented by an elementary matrix. Let $E_1, E_2, \ldots, E_k$ be the elementary matrices corresponding to the sequence of elementary row operations that transform A into I_n. Then, we can write:

$$E_k E_{k-1} \ldots E_2 E_1 A = I_n$$

Since each elementary matrix is invertible, and product of invertible matrices is invertible.

$$\begin{aligned}(E_k E_{k-1} \ldots E_2 E_1)^{-1}(E_k E_{k-1} \ldots E_2 E_1 A) &= (E_k E_{k-1} \ldots E_2 E_1)^{-1} I_n \\ \implies I_n A &= E_1^{-1} E_2^{-1} \ldots E_{k-1}^{-1} E_k^{-1} \\ \implies A &= E_1^{-1} E_2^{-1} \ldots E_{k-1}^{-1} E_k^{-1}\end{aligned}$$

Since the inverse of an elementary matrix is also an elementary matrix, so A is expressed as product of elementary matrices.

Problem 12: Let A and B be two square matrices of the same size. If $AB = I$, then prove that $B = A^{-1}$.

Solution

Given that $AB = I$, then $\det(AB) = \det(A)\det(B) = 1$ implies that $\det(A)$ is non-zero and A is invertible.

Now, multiply $AB = I$ on the left by A^{-1} (which exists because A is invertible):

$$\begin{aligned}A^{-1}(AB) &= A^{-1} I \\ \implies (A^{-1}A)B &= A^{-1} \\ \implies IB &= A^{-1} \\ \implies B &= A^{-1}\end{aligned}$$

Therefore, if $AB = I$, then $B = A^{-1}$.

Problem 13: Prove the following identity without evaluating the determinant:

$$\begin{vmatrix} a & b & a+b+c \\ d & e & d+e+f \\ g & h & g+h+i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Solution

Subtract the first column from the third column:

$$\begin{vmatrix} a & b & a+b+c \\ d & e & d+e+f \\ g & h & g+h+i \end{vmatrix} = \begin{vmatrix} a & b & b+c \\ d & e & e+f \\ g & h & h+i \end{vmatrix}$$

Subtract the second column from the third column:

$$\begin{vmatrix} a & b & b+c \\ d & e & e+f \\ g & h & h+i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Thus, we have shown that:

$$\begin{vmatrix} a & b & a+b+c \\ d & e & d+e+f \\ g & h & g+h+i \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Problem 14: Let A and B be two square matrices of the same size. If $BA = I$, then prove that $B = A^{-1}$.

Solution

Same as Problem 12.

Problem 15: Prove the following identity without evaluating the determinant:

$$\begin{vmatrix} a+b & a-b & c \\ d+e & d-e & f \\ g+h & g-h & i \end{vmatrix} = -2 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Solution

Same as Problem 13 with few extra steps where we take 2 common from a column and perform a switching as well.